

Alpaca 30-Day Trading Challenge

Strategy Rundown — Plain English Edition

Challenge period: **April 13 – May 13, 2026** | Capital at risk: **\$10,000** | Full account: **\$50,000** (rest stays safe)

1. The Big Picture — What Are We Doing?

We are running an automated stock-trading bot on a **paper trading** account (real market prices, but no real money at risk). The bot watches **16 stocks** every hour during market hours and decides whether to buy, hold, or sell based on a strict set of rules. Only **\$10,000** of the \$50,000 account is used — the rest sits untouched. The bot is optimised for competition: larger position sizes, more setups, and leveraged ETFs to maximise returns over the 30-day window.

2. The Watchlist — Stocks We Track

The bot monitors **16 tickers** for trade opportunities — expanded from 7 to include the highest-momentum names and leveraged ETFs for maximum upside:

Ticker	What It Is	Why It's Here
TQQQ	3× Nasdaq ETF (leveraged)	Triples every QQQ move — biggest upside in bull runs
SOXL	3× Semiconductor ETF (leveraged)	Semiconductor boom exposure at 3× speed
NVDA	NVIDIA — AI chip leader	Strong trend, clear levels, high volume
TSLA	Tesla — highest-beta megacap	Moves fast and far; ideal for breakout trades
META	Meta Platforms — AI advertising leader	AI tailwind + strong trend momentum
AMD	Advanced Micro Devices	Tracks NVDA closely, often cheaper to enter
MSFT	Microsoft — cloud & AI	Stable uptrend, large moves on catalyst days
AAPL	Apple	High liquidity, reacts to broad market shifts
AMZN	Amazon — cloud + retail	Trending stock with large move potential
GOOGL	Alphabet — Google / AI	AI search cycle tailwind
PLTR	Palantir — AI data analytics	High-beta, momentum-driven breakouts
COIN	Coinbase — crypto exchange	Volatile, high-beta when risk-on
XLF	Financial sector ETF	Bank earnings week = direct catalyst
XLE	Energy sector ETF	Rotation & commodity-driven moves

XLK	Technology sector ETF	Broad tech exposure
SPY	S&P 500 index ETF	Market health check only — never traded
QQQ	Nasdaq 100 index ETF	Market health check only — never traded

Green rows = leveraged ETFs (3× amplification). Grey rows = market health indicators only, never traded. The scanner auto-discovers the best setup across all 16 symbols on each trading day.

3. The Three Types of Trades We Use

The bot uses three setups — all require the stock to **prove itself** before we enter. No guessing, no chasing.

Trade Type	Plain-English Idea	Best Symbols
Breakout Trade	A stock keeps hitting the same ceiling price over and over. When it finally punches through that ceiling on high volume, it often runs higher fast. We buy that breakout moment.	XLK, TQQQ, SOXL, TSLA, NVDA
Pullback Trade	A stock had a big breakout day, then cooled off a little. Instead of chasing it at the peak, we wait for the dip and buy at a better price — lower risk, same upside.	NVDA, AMD, META, MSFT, AAPL
Momentum Breakout (NEW)	Stock hits a new 20-day high on 1.5× normal volume. Pure price strength. The market is already voting with money that this stock wants to go higher — we join it.	TQQQ, SOXL, TSLA, PLTR, COIN, NVDA

Why This Strategy Wins the Challenge

- **4× bigger positions** — 2% risk per trade (\$200) vs 0.5% before; same win rate = 4× the profit
- **Leveraged ETFs (TQQQ, SOXL)** — triple the Nasdaq/semiconductor move; huge upside in trending markets
- **16 stocks watched** — more opportunities found every day by the auto-scanner
- **Hourly checks** — bot runs 7× per day instead of 5× to catch entries at the right moment
- **Three setup types** — breakout, pullback, and momentum; always a setup somewhere in the market
- **Tighter trailing stop** — locks in gains at +5% (was +10%) so profits aren't given back

4. Current Active Setups — What We're Watching Right Now

Two trades are pre-loaded and ready to trigger:

	XLK — Breakout Setup	NVDA — Pullback Setup
Idea	Buy if price breaks above \$51.30	Buy on a pullback to \$184–\$185.50

Why	Banks coiling below resistance; earnings week = catalyst	Broke above 50-day average on strong volume; waiting for a dip to re-enter
Entry	\$51.30	\$184.50
Stop	\$50.50 (lose ≈\$0.80/share)	\$180.50 (lose ≈\$4/share)
Target 1	\$52.50 (+2.3%)	\$192.00 (+4.1%)
Target 2	\$54.00 (+5.3%)	\$197.00 (+6.8%)
Score	71 / 100	76 / 100

Scores are based on the 5-factor framework (Section 6). A score of 65+ is required to even watch a stock; 80+ triggers immediate entry.

5. How a Trade Works — Step by Step

Step 1 — Market health check: Before any buy, the bot checks SPY (S&P 500). If the market is in a downtrend, position sizes are cut in half or the bot sits out entirely.

Step 2 — Entry trigger: The bot checks whether the stock's live price has hit the target entry zone (breakout above a level, or a pullback into a support zone).

Step 3 — Volume check: Volume must be at least 40% of the 20-day average. Low-volume moves are ignored — they don't have enough conviction behind them.

Step 4 — Risk/reward gate: The potential profit must be at least **1.5× the potential loss**. If the math doesn't work out, no trade.

Step 5 — Position sizing: The bot risks no more than **0.5% of \$10,000 (\$50) per trade**. It calculates exactly how many shares to buy so that if the stop-loss is hit, the loss is capped at \$50.

Step 6 — Managing the trade: See Section 6 below.

6. Exit Rules — How We Get Out of a Trade

There are four ways a trade ends:

Exit Type	What Happens	When
Hard Stop Loss	Sell immediately at a small loss	Price drops to the stop level (set before entry)
Target 1 (T1)	Sell half the position, lock in profit	Price rises to the first profit target

Trailing Stop	Dynamic stop that rises with the price, then triggers if price falls back	Activates when gain reaches +10%; trails 5% below the highest price
End of Day	All open orders expire (not positions)	Market close — unfilled orders are cancelled automatically

7. How Stocks Are Scored — The 5-Factor Model

Every potential trade is scored out of 100 points across five categories. The bot only acts on scores that meet the minimum threshold.

Factor	Max Points	What It Checks
Regime Alignment	20	Is the overall market (SPY) healthy?
Setup Clarity	20	Is the chart pattern clean and obvious?
Market Confirmation	20	Are sector ETFs and volume confirming the move?
Risk/Reward Quality	20	Is the profit potential at least 1.5× the risk?
Event Cleanliness	20	No earnings, Fed announcements, or news that could blow up the trade?
TOTAL	100	

Score Range	Decision	Meaning
80–100	EXECUTE ✓	All signals aligned — enter the trade
65–79	WATCH →	Promising but not perfect — monitor closely
Below 65	PASS ✗	Too many question marks — skip it

8. Hard Rules — Lines the Bot Will Never Cross

These are non-negotiable safety limits built into the code:

Rule	Limit	Notes
Max loss per trade	\$200	2% of \$10,000 — 4× larger than before for competition
Max loss per day	\$400	4% of \$10,000 — more room to trade aggressively
Max total open risk	\$400	4% of \$10,000 across all open positions
Max trades per day	5	Up from 3 — more opportunities captured

Min profit/risk ratio	2.0×	Must make at least \$2.00 for every \$1.00 risked
Trailing stop trigger	+5%	Activates at +5% gain (was +10%) — locks in profits sooner
Capital cap	\$10,000	Never touches the other \$40,000 in the account

9. Automation Schedule — When Does the Bot Run?

Time (ET)	What Happens
10:00 AM	First check — 30 min after open; avoids the chaotic opening surge
11:00 AM	Second check — looks for new setups and manages open positions
12:00 PM	Midday check — captures late morning momentum moves
1:00 PM	Afternoon check — updates trailing stops on open positions
2:00 PM	Mid-afternoon check — highest probability entry window
3:00 PM	Pre-close check — tightens stops ahead of closing volatility
4:00 PM	Final check — market close; open orders expire automatically
After 4 PM	Market closed — bot skips instantly and waits for next day

Bot runs every hour on weekdays (Mon–Fri): 7 checks per day vs 5 before. Weekend and holiday runs are skipped automatically.

10. The Dashboard — Tracking Progress

A live dashboard runs locally at <http://localhost:5050>. Double-click **Open Dashboard.command** in the folder to launch it.

- Total P&L for the challenge (vs. \$10,000 starting capital)
- Today's P&L
- Current open positions with live prices
- Active triggers — what the bot is watching and at what price
- Full trade history with entry/exit prices and profit/loss per trade
- Daily P&L bar chart — visual trend over the 30 days

The dashboard auto-refreshes every 2 minutes. You can also access raw data at <http://localhost:5050/api/data>

11. Files in Your Folder — What Does What

File	What It Does
<i>bot.py</i>	The main trading bot — runs every 2 hours automatically
<i>dashboard.py</i>	The web dashboard server (localhost:5050)
<i>Open Dashboard.command</i>	Double-click to launch the dashboard in your browser
<i>run_bot.sh</i>	Shell script that cron uses to run the bot
<i>strategy.json</i>	Current active trade setups and bot state
<i>performance.json</i>	Running P&L totals and daily breakdown
<i>trade_log.json</i>	Every trade the bot has made (entry, exit, profit/loss)
<i>bot.log</i>	Full bot activity log — useful for debugging
<i>.env</i>	Your Alpaca API credentials (keep this private)
<i>Skill.md</i>	The detailed trading rules the bot follows
<i>Strategy_Rundown.pdf</i>	This document

Remember: This is a paper trading challenge. No real money is at risk. The bot is fully automated — you don't need to do anything. Just open the dashboard whenever you want to check in on progress.